

Project Statement: Identifying Optimal Passing Patterns Using Temporal Subgraph Isomorphism for Football Teams

Background

In football, passing patterns can determine the effectiveness of a team's play. In football, understanding the passing patterns between players can provide valuable insights into team dynamics and strategies. To analyze these patterns, we can model the players and their passes as a dynamic graph, where the passing connections between players change over time. Here's how it works and how it connects to the concept of dynamic graph isomorphism:

1. Graph Representation of a Team:

Each player on the team is represented as a node (or vertex) in the graph. A pass between two players is represented as an edge connecting those two nodes. For example, if Player A passes to Player B, there is an edge from A to B. The weight or thickness of the edge can represent factors like the frequency or success rate of passes.

2. Dynamic Nature of the Graph:

In a football game, passing patterns evolve over time based on the game's flow, player positions, and strategies. A passing network may look very different at the beginning of a match compared to the end. This temporal evolution is what makes it a dynamic graph, where edges can appear, disappear, or change weight as passes happen in real-time.

3. Graph Isomorphism in Passing Patterns:

Graph isomorphism is the concept of determining if two graphs are structurally identical, even if the node labels (i.e., player identities) differ. In football, this means analyzing if one passing pattern or team structure resembles another, even when the specific players involved differ.

4. Mapping Dynamic Graph Isomorphism to Passing Patterns:

Dynamic graph isomorphism takes this a step further by comparing the passing structures across different time intervals or games. For example:

- We could check if a team's passing strategy at the beginning of a match is "structurally similar" to a strategy used in another match or later in the same game, regardless of who exactly is involved.

- If a team consistently has certain positions or formations in passing patterns, we might find recurring isomorphic structures. For example, a triangular passing setup among three players may appear in multiple games or even with different players.

5. Benefits of Understanding Dynamic Graph Isomorphism in Football:

By mapping passing patterns to dynamic graph isomorphisms, coaches and analysts can gain insights such as:

- Identifying signature playing styles or recurring structures that give the team a strategic advantage.
- Detecting when and how the team's structure changes under different pressures, like aggressive pressing from the opponent.
- Adjusting in-game strategies by identifying patterns that resemble successful setups from past matches.

In essence, mapping passing patterns in football to dynamic graph isomorphisms allows us to detect and analyze evolving team strategies in a structured way. This approach can be a powerful tool for coaches and players to understand team connectivity and adaptability throughout the game.

A recent technique proposed by [Oberoi et al. \(2023\)](#) focuses on detecting dynamic subgraph isomorphism patterns in temporal graphs. This technique can be applied to understand the movement and passing sequences of players in football by treating player interactions as dynamic graphs. The code of the paper is available [here](#). This project will explore how passing patterns evolve and which strategies are most effective using this novel approach. Another paper by [Bourgeois et al. \(2023\)](#) has used this concept for identifying passes of basketball team

What to do

1. You will understand the primary steps of the two research papers.
2. You will write code that will transform the dataset of passes available for football matches here so that it can be processed by given code. Dataset available [here](#)
3. You will provide analysis of the two teams similar to what is shown in Fig. 3 and Fig. 4 of Bourgeois paper.

The code provided by Oberoi expects the input in the following format.

On the first line there must be the number of nodes; subsequent lines will contain the node attributes, one node per line, preceded by the node id; node ids must be in the range from 0 to the number of nodes - 1. Then, for each node there is the number of edges coming out of the node, followed by a line for each edge containing the ids of the edge ends and the edge attribute. Blank lines, and lines starting with #, are ignored. An example file, where both node and edge attributes are ints, could be the following:

```
# Number of nodes
3
```

```

# Node attributes
0 27
1 42
2 13

# Edges coming out of node 0\n
2
0 1 24
0 2 73

# Edges coming out of node 1
1
1 3 66

# Edges coming out of node 2
0

```

Project Steps:

0. Present understanding of both papers and code, input and out. 25% of weight Due 15th of Nov

1. Data Preparation: 40%

dataset of football passes, where each pass represents a directed edge between players (nodes). Ensure the data includes timestamps for every pass made during a match. Create a dynamic graph where nodes represent players, and edges represent the passing actions over time. Use the snapshot-based model to capture these interactions at different time intervals (i.e., sequence of passes within specific time windows).

2. Adapt the Technique: 5%

Adapt the temporal subgraph isomorphism technique provided in Oberoi et al. (2023). This will involve utilizing the code provided to detect dynamic subgraph patterns across different match snapshots.

3. Identify Key Passing Patterns: 10%

identify and analyze frequently occurring isomorphic patterns in the passing dataset. These patterns should highlight effective passing strategies.

4. Comparison of Patterns: 20%

Compare the detected patterns of the two team